FESP OVERVIEW (rev Fall 2018)

Engineers solve problems, and the six departments that make up University of Illinois at Chicago's College of Engineering are working to help their students join those problem-solvers. At UIC, we know that the change from high school to college is a big one, and we understand that it can be difficult to adjust to the rigors of a university engineering program.

We have developed a methodology to put freshmen on the path to early success as well as to welcome them to our community. It is called the *Freshman Engineering Success Program (FESP)*. The *Freshman Engineering Success Program* facilitates this transition for our students, inspires them about their major, and provides them with the tools to be successful in their freshman year and beyond. It is the first step in becoming part of the engineering community at UIC and identifying as an engineer. It also rewards them with a guaranteed paid internship at the end of the year.

Now in its sixth year, FESP was first added in 2013 to the ENGR100 Engineering Orientation course taken by students newly enrolled in the College of Engineering. FESP is now fully incorporated into ENGR100 and includes the following activities; an alumni panel discussion, two team-based projects, a pre-advising session, as well as many opportunities throughout the semester to interact with upper-class engineering students and faculty in the students' majors. Next year, FESP will be expanded to include transfer students that also are required to take ENGR100.

The FESP activities are designed to provide different engineering-related experiences to the students. The alumni panel discussions are made up of recent UIC alumni from all the engineering departments. The students ask questions to the panel of engineering professionals and gain a better understanding of what it means to be an engineer working in industry, in the government, or in academia.

The goal of the first team project is for the students to gain an understanding of their major by conducting an in-depth analysis of the curriculum. In addition, they interview at least eight upper-classmen, thus establishing relationships with other students in their major.

The second team project provides a discipline-specific, hands-on project that introduces the students to engineering problems and builds an engineering sense while using their existing math and science skills.

The two team-based projects foster communication with engineering faculty and upper class students. Since freshman students are busy taking their pre-requisite courses, there is not much interaction with their department or any of the faculty and students in their major. The students get a chance to be mentored by engineering faculty and upper-class student teaching aides while working on their projects. This provides invaluable experiences at collaborating with engineering experts and students in their major and helps break down the barriers that can exist between students and faculty.

In addition, a pre-advising session is held prior to the mandatory, faculty advising appointment. The objective is to provide a follow-up to the first project and to offer further advice on specific issues related to the major. In addition, the students meet department staff who will be helping them coordinate their education while at UIC.

Students who successfully complete the FESP requirements and maintain a GPA of 3.2 or greater during the first and second semester qualify for the Guaranteed Paid Internship Program (GPIP) during the summer that follows their freshman year. The freshmen are placed in engineering firms in the Chicago area or in a research laboratory at UIC, where they are exposed to real world engineering practices, build relationships with industry professionals and college professors, and gain valuable experience. In its first two years, more than 200 freshmen have participated in the program and worked summer internships.